

Dr. Kuldeep S. Yadav

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Technical Summary

Experienced researcher with 6+ years in deep learning, computer vision, and affective computing. Skilled in multimodal signal processing (image, video, audio, EEG/EMG). Strong publication record with 7 patents and 18 SCI/SCIE journals.

Experience

- Apr 2023 – **Scientist**, CSIR–Fourth Paradigm Institute, Bengaluru, India
Present Leading projects in AI-based cybersecurity, data science, and anomaly detection.
- Apr 2023 – **Assistant Professor**, Academy of Scientific and Innovative Research(AcSIR)
Present Teaching and Mentoring PhD Students
- Apr 2023 – **Postdoctoral Fellow**, IIT Delhi – EE Dept., India
Apr 2025 Research on multimodal suspiciousness estimation (image, video, audio, EEG/EMG).
- Aug 2022 – **Young Professional**, Department of Telecommunication, Govt. of India, India
Feb 2023 Research in 6G, DLT, blockchain, and last-mile connectivity under USOF.
- Mar 2019 – **Research Fellow**, IMPRINT-II DST, SERB, India
Jul 2022 Deep learning for image/video processing, particularly for object localization and gesticulated character translation using MATLAB and Python.

Education

- 2019 – 2023 **Ph.D., ECE**, NIT Silchar, India
Specialization: Image Processing & Computer Vision
- 2016 – 2018 **M.Tech, ECE**, LNMIIT Jaipur, India
Specialization: Image Processing — CGPA: 6.9
- 2012 – 2016 **B.Tech, ECE**, Dr. B R Ambedkar University, Agra, India
74.9%

Technical Skills

- Programming Python, MATLAB, C/C++, LaTeX, PyTorch, TensorFlow
- Tools Overleaf, MS Office, GUI Design
- Domains Pattern Recognition, Machine Learning, Deep Learning, Signal & Image Processing

Research Profile

- Patents 7 (including USE-Riskometer)
- Publications 18 SCI/SCIE Journals, 12 International Conferences

Research Interests

Topics Computer Vision, Affective Computing, Hand Gesture Recognition, Object Detection, Multimodal Analytics

Languages

English Fluent

Hindi Native

Projects

IMPRINT-II Development of a prototype of a disabled-friendly automatic virtual text-entry keyboard (DST, SERB) interface system under practical environmental conditions.

Funding: 9 million INR

Agency: IMPRINT-II, DST, SERB, India

Patents

India **USE-Riskometer:** A system and method for estimating suspiciousness in an environment. *Indian Patent App. No. 202511059929*

India **FER20E:** A system for facial expression recognition using attention based vision transformer. *Indian Patent App. No. 202411029187*

India **Dynamic gesture character recognition system:** System for recognizing hand-gesticulated characters. *Indian Patent App. No. 202231036486*

Germany **Speed-invariant technique system:** A system to identify and remove self co-articulations in alphanumeric characters. *German Utility Model No. 20220101980*

Germany **Gesture object detection & tracking:** Deep learning-based system for gesture object recognition. *German Utility Model No. 202022105250*

Germany **Facial Expression Recognition:** Real-time system for recognizing facial expressions. *German Utility Model No. 202022105196*

Germany **Speech-based heart rate estimator:** System for estimating discrete heart rate values from speech signals. *German Utility Model No. 202022104091*